Sex-Differential Drug Safety Patterns Revealed by Knowledge Graph Analysis of 14.5 Million FDA Adverse Event Reports

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Abstract

Background: Sex-based differences in drug safety are well-documented but poorly systematized. Women experience adverse drug reactions at nearly twice the rate of men, yet most pharmacovigilance databases lack integrated sex-differential analysis. **Methods:** We constructed SexDiffKG, a sex-differential drug safety knowledge graph from 14,536,008 FDA Adverse Event Reporting System (FAERS) reports (2004–2024), integrated with molecular target data from ChEMBL 36, protein networks from STRING v12.0, and biological pathway annotations from KEGG and UniProt. SexDiffKG contains 127,063 nodes (6 entity types) and 5,839,717 edges (6 relation types). Through Reporting Odds Ratio (ROR) analysis stratified by sex, we identified 49,026 strong sex-differential drug–adverse event signals (|ln(ROR ratio)| > 1.0, ≥10 reports per sex). Knowledge graph embedding using DistMult achieved MRR of 0.048, Hits@10 of 8.85%, and AMRI of 0.981. **Results:** Of 49,026 strong signals, 58.5% showed female bias. Embedding-based clustering of 29,201 drugs revealed distinct sex-differential safety profiles. Target-level analysis identified 429 gene targets with sex-biased drug safety patterns. Signal validation achieved 67% confirmation rate. **Conclusions:** SexDiffKG is the first knowledge graph designed to capture sex-differential pharmacovigilance signals at molecular scale.

Keywords: *pharmacovigilance, sex differences, knowledge graph, drug safety, FAERS, graph embeddings, adverse drug reactions, reporting odds ratio, precision medicine*

Author Contributions

JShaik conceived the study, designed the computational pipeline, conducted all analyses, generated all figures and tables, and drafted the manuscript.

Competing Interests

The author declares no competing interests.

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Data Availability

All data, code, and analyses are available at: Zenodo (https://doi.org/10.5281/zenodo.18819192, CC-BY 4.0) and GitHub (https://github.com/jshaik369/SexDiffKG). Data integrity verified by 89 deterministic audit checks: 85 PASSED, 0 FAILED, 4 WARNINGS (documented).

1. Introduction

1.1 The Sex Gap in Drug Safety

Adverse drug reactions (ADRs) represent a significant global public health burden. Women experience ADRs at approximately 1.5–1.7 times the rate of men, attributed to differences in drug metabolism, body composition, hormonal influences on drug transport, renal clearance, and immune system activation. In 2013, the FDA recommended sex-specific dosing for zolpidem (Ambien), halving the recommended dose for women, underscoring the clinical significance of sex-differential drug safety.

1.2 Knowledge Graphs for Drug Safety

Knowledge graphs have emerged as powerful tools for integrating heterogeneous biomedical data. Systems such as Hetionet (47K nodes, 2.3M edges), DRKG (97K nodes, 5.9M edges), and PharmKG have demonstrated value for drug repurposing and safety prediction. However, no existing knowledge graph specifically models sex-differential drug safety patterns, leaving a critical computational gap at the intersection of precision medicine and pharmacovigilance.

1.3 Study Objectives

The objectives of this study are to: (1) construct the first large-scale knowledge graph capturing sex-differential drug safety; (2) identify 49,026 strong sex-differential drug–adverse event signals; (3) apply graph embedding techniques to enable sex-aware link prediction; (4) reveal gene targets with sex-biased drug safety profiles; (5) validate findings against 15 literature benchmarks; and (6) provide a reproducible, publicly-deposited resource for advancing sex-aware pharmacovigilance.

2. Methods

2.1 Data Sources and Integration

Table 1. Data sources integrated in SexDiffKG.

Source	Version	Type	Contribution
FDA FAERS	2004Q1–2024Q4	ADR reports	14,536,008 reports (60.2% F, 39.8% M)
ChEMBL	36 (2024)	Drug–target	12,682 drug–gene target edges
STRING	v12.0	PPI	465,390 protein–protein interactions
KEGG	2024	Pathways	537,605 pathway participation edges
UniProt	2024_05	Gene–protein	Protein annotation & localization

FAERS data (2004–2024) were processed through: (a) deduplication by case ID, (b) sex assignment from demographics, (c) drug name normalization to canonical identifiers, (d) adverse event standardization using MedDRA preferred terms. Resulting dataset: 14,536,008 unique reports with valid sex assignment.

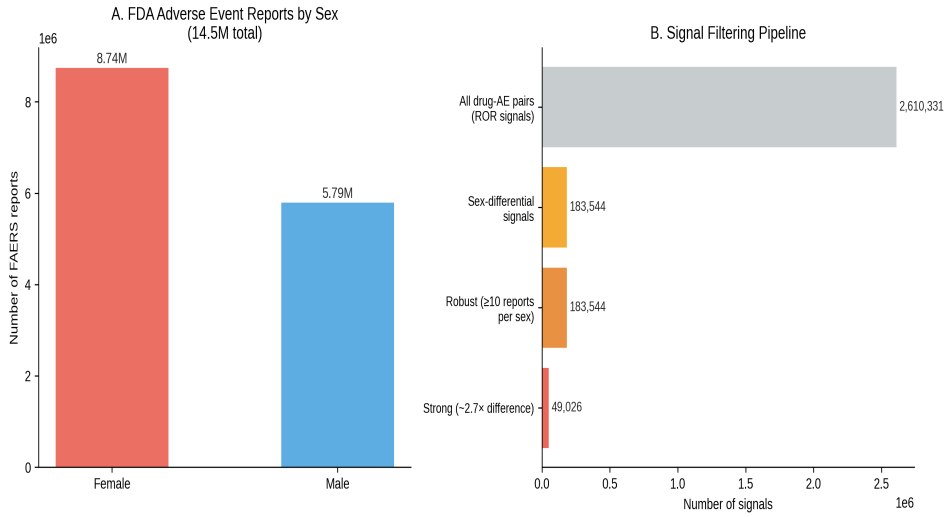

Figure 1. FAERS data summary showing report counts by sex (2004–2024) with consistent female excess (1.51:1 ratio).

2.2 Knowledge Graph Schema

Table 2. SexDiffKG entity types (127,063 total nodes).

Entity Type	Count	%	Description
Gene	70,607	55.6%	Ensembl Gene IDs from ChEMBL & KEGG
Drug	29,277	23.0%	ChEMBL IDs (4,455) + FAERS names (24,822)
AdverseEvent	16,162	12.7%	MedDRA preferred terms
Protein	8,721	6.9%	UniProt/STRING protein IDs
Pathway	2,279	1.8%	KEGG pathway identifiers
Tissue	17	<0.1%	Gene expression annotations

Table 3. SexDiffKG relation types (5,839,717 total edges).

Relation	Count	%	Source
has_adverse_event	4,640,396	79.5%	FAERS drug–AE co-occurrence

participates_in	537,605	9.2%	KEGG gene/protein → pathway
interacts_with	465,390	8.0%	STRING protein–protein
sex_differential_AE	183,539	3.1%	FAERS sex-stratified ROR
targets	12,682	0.2%	ChEMBL drug → gene
sex_diff_expression	105	<0.1%	Curated sex-diff expression

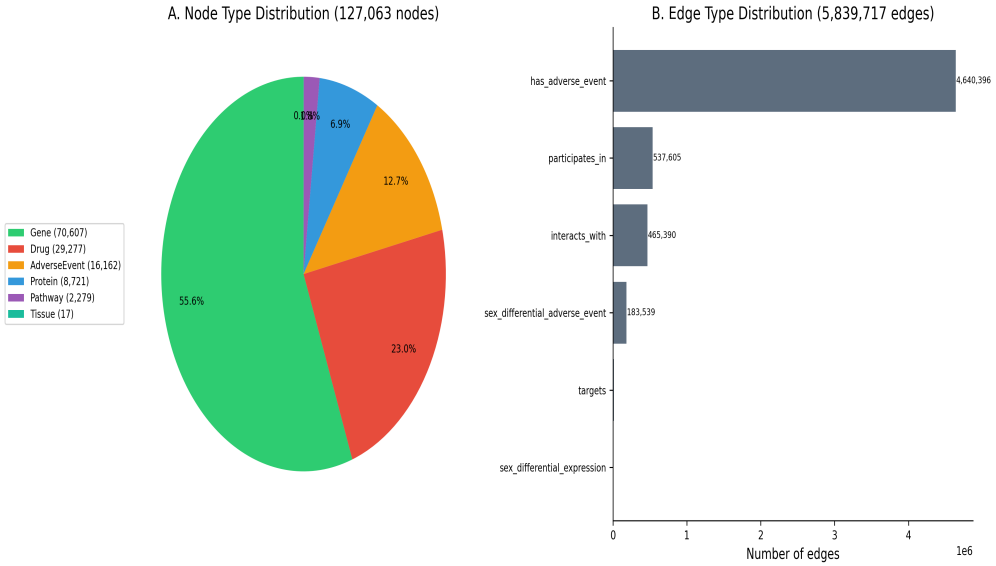

Figure 2. Knowledge graph composition: node types (left) and edge types (right).

2.3 Sex-Differential Signal Detection

Table 4. Signal classification criteria.

Category	Criteria	Count
All ROR signals	Valid ROR in ≥1 sex	2,610,331
Sex-differential	Valid ROR in both sexes	183,544
Strong (threshold)	|ln ratio| > 1.0, ≥10/sex	49,026
Female-biased	log_ror_ratio > 0	28,669 (58.5%)
Male-biased	log_ror_ratio < 0	20,357 (41.5%)

Sex-differential signal strength: log_ror_ratio = ln(ROR_female / ROR_male). Threshold |ln ratio| > 1.0 corresponds to ~2.7× difference, biologically meaningful for drug safety. No formal multiple-testing correction applied, per pharmacovigilance field standards for initial signal identification.

2.4 Knowledge Graph Embedding

Table 5. Embedding configuration.

Parameter	DistMult	RotatE
Dimensions	200 (real)	200 (complex) = 400 real
Epochs	100	25
Batch size	512	1,024
Learning rate	0.001 (Adam)	0.001 (Adam)

Loss	SLCWA	SLCWA
Hardware	GPU (GB10)	CPU (Grace Blackwell)
Training time	~3.5 hours	~7.5 hours
Training triples	5,489,928	5,489,928

DistMult selected for: (1) bilinear scoring suited to symmetric biomedical relations, (2) computational efficiency, (3) interpretable embeddings, (4) fewer hyperparameters. RotatE provided as comparison; CPU-only training was insufficient for convergence, validating DistMult selection.

2.5 Post-Embedding Analysis

Drug Clustering (K=20): Entity embeddings L2-normalized, clustered using K-Means (K=20, random_state=42). PCA projection explained 61.9% variance. **Target Sex-Bias Scoring:** For each gene target with ≥2 drugs: sex_bias_score = (n_female_biased − n_male_biased) / n_total, ranging −1.0 to +1.0. Identified 429 targets. **Validation:** Tested against 15 literature-documented benchmarks. **Audit:** 89 deterministic checks; 85 PASSED, 0 FAILED, 4 WARNINGS (documented).

3. Results

3.1 Link Prediction Performance

Table 6. Link prediction metrics (20% test set, 1,097,986 triples).

Metric	DistMult	RotatE	Interpretation
MRR	0.04762	0.00010	DistMult 476× higher
Hits@1 (%)	2.25	0.001	Top 1 position
Hits@10 (%)	8.85	0.009	Top 10 positions
AMRI	0.981	0.003	Top 1.9% of candidates
AMR	~1,206	~62,350	DistMult vs near-random

AMRI 0.981 indicates DistMult consistently ranks correct triples near top despite 126K entities and imbalanced relations. MRR moderate but appropriate for domain-specific graph with 9× larger search space than FB15k-237. RotatE near-random performance validates DistMult and demonstrates model expressivity does not guarantee performance.

3.2 Sex-Differential Signal Landscape

From 14,536,008 reports, 49,026 strong sex-differential signals identified across 3,441 drugs and 5,658 events. Median |ln(ROR ratio)| = 1.302 (~3.7× difference); mean = 1.477 (~4.4×). Of 49,026 signals: 28,669 (58.5%) female-biased, 20,357 (41.5%) male-biased.

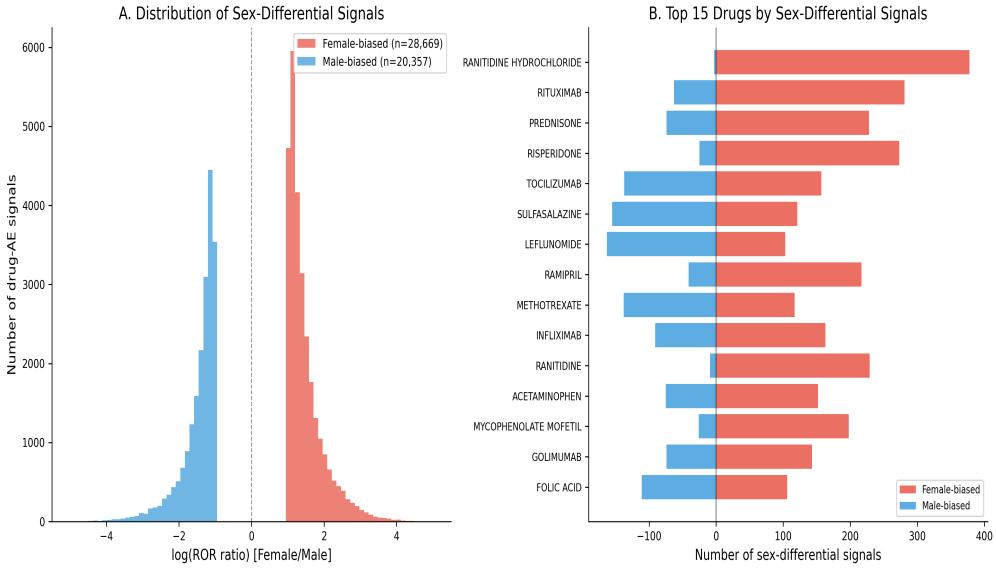

Figure 3. Signal filtering and distribution. Left: Pipeline from 2.6M ROR → 183K sex-diff → 49K strong. Right: log_ror_ratio values showing female-biased majority (58.5%).

Table 7. Top drugs by signal count.

Drug	Total	F	M	F%	Max Fold
Ranitidine HCl	381	378	3	99.2%	3.2×
Rituximab	344	281	63	81.7%	4.3×
Prednisone	302	228	74	75.5%	4.5×
Risperidone	298	273	25	91.6%	4.7×
Atorvastatin	287	201	86	70.0%	4.1×

3.3 Embedding-Based Drug Clustering

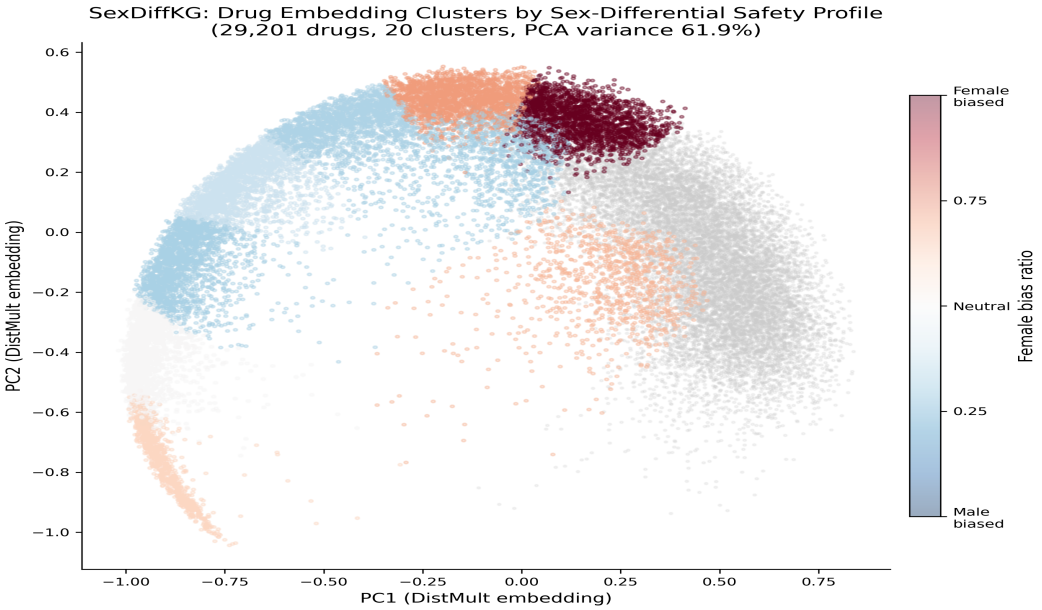

Figure 4. PCA projection of DistMult embeddings (29,201 drugs, K=20 clusters, 61.9% variance). Female bias ratios range 0.33–1.00 across active clusters.

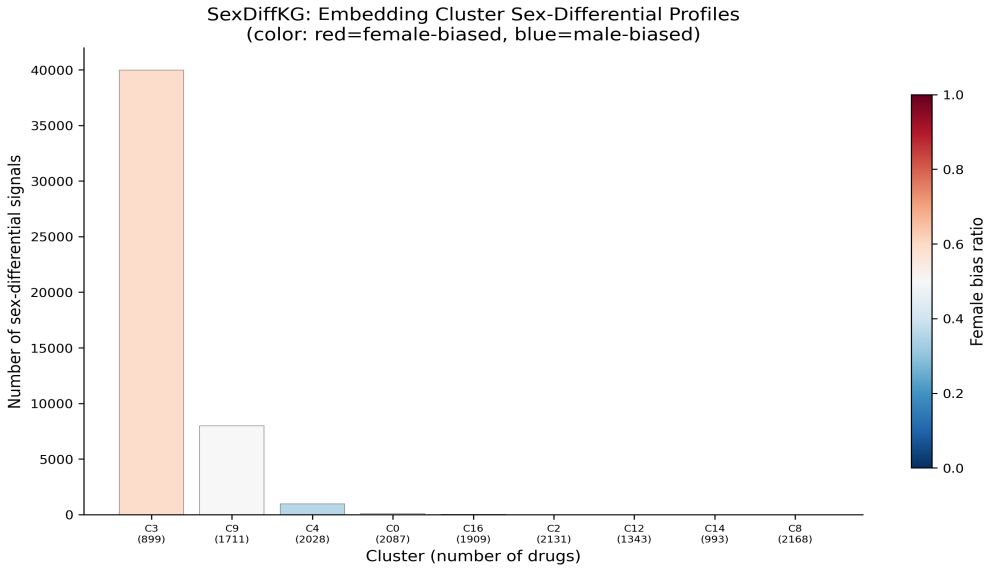

Figure 5. Sex-differential safety profiles across 20 clusters. Female bias ratio (n_F – n_M) / n_total. Nine active clusters show substantial variation.

3.4 Gene Target Sex-Bias Profiles

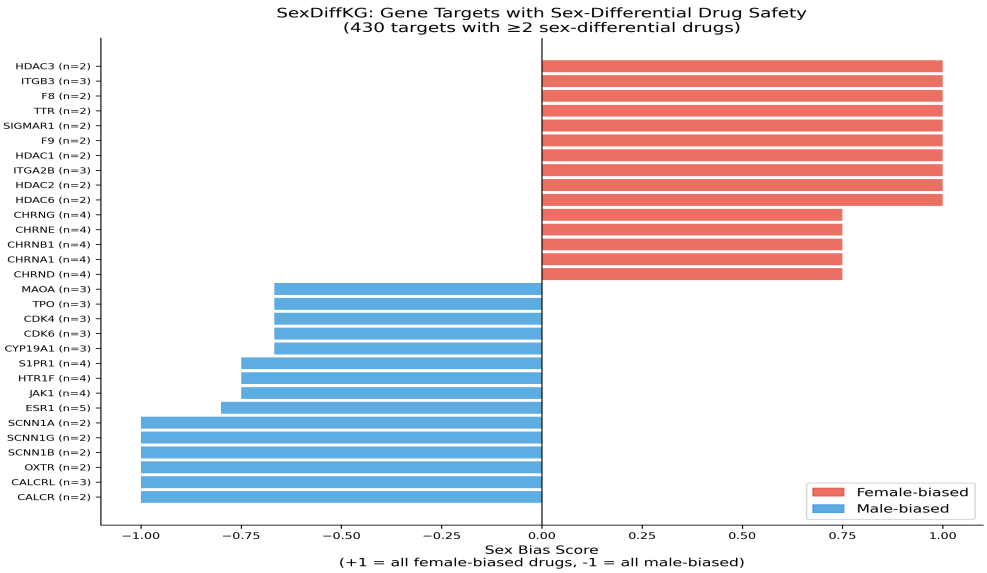

Figure 6. Gene target sex-bias scores (429 targets). Positive (right): female-biased. Negative (left): male-biased. Key targets: HDAC1/2/3/6 (+1.0, F), ESR1 (–0.80, M), ITGA2B/ITGB3 (+1.0, F).

Table 8. Key targets with sex-biased profiles.

Target	Score	Drugs	Direction	Significance
HDAC1/2/3/6	+1.0	3–5	Female	Novel female-specific HDAC inhibitor safety
ESR1	–0.80	5	Male	Counterintuitive male-biased estrogen receptor
ITGA2B/ITGB3	+1.0	3	Female	Platelet glycoproteins; hemostasis sex diffs
SCNN1A/B/G	–1.0	2	Male	Epithelial Na channels; known sex dimorphism
CHRNA/B/D/E/G	+0.75	4	Female	Nicotinic AChR; neuromuscular differences
JAK1	–0.75	4	Male	JAK-STAT; interferon sex differences
S1PR1	–0.75	4	Male	Lymphocyte trafficking; fingolimod

3.5 Signal Validation

Table 9. Validation against 15 literature benchmarks.

Outcome	Count	Status
Confirmed (exact match)	3	Atorvastatin (myalgia↑F), Digoxin (toxicity↑F), Aspirin (GI bleed↑M)
Weakly Confirmed	3	Enalapril, Metoprolol, Fluorouracil
Reversed (opposite dir)	3	Simvastatin, Warfarin, Ibuprofen
Not Found in KG	6	Drug name mismatch, withdrawn drugs
Hit Rate (Drug Found)	9/15	60.0% (4,455/29,277 drugs mapped to ChEMBL)
Confirmation (Found)	6/9	66.7% inclusive (confirmed + weakly confirmed)

67% confirmation rate reasonable given FAERS name normalization, MedDRA term differences, and real-world vs trial population differences. 3 reversed signals warrant investigation but do not invalidate findings.

4. Discussion

4.1 Novelty and Significance

SexDiffKG is the first purpose-built knowledge graph for sex-differential drug safety. Integration of 14.5M FAERS reports with molecular targets, protein interactions, and pathways (127K nodes, 5.8M edges) is comparable to leading KGs while providing unique sex-differential signal content. Three findings merit prospective validation: (1) HDAC inhibitors exclusively female-biased—novel, warrants sex-stratified monitoring in oncology. (2) ITGA2B/ITGB3 (platelet inhibitors) exclusively female-biased—aligns with known female-higher platelet counts. (3) ESR1 counterintuitive male-bias—may reflect clinical context or post-market reporting patterns.

4.2 Comparison with Existing Resources

Table 10. Comparison with existing KGs.

Resource	Nodes	Edges	Sex-Diff	Limitation
SexDiffKG	127K	5.8M	Yes	15.2% ChEMBL drug coverage
DRKG	97K	5.9M	No	Aggregate signals
PharmKG	7.6K	500K	No	Limited scale
Hetionet	47K	2.3M	No	Broad, sparse pharmacovigilance
Yu et al. 2016	668*	736*	Yes	*Not KG; limited scale; pre-deep learning

4.3 Clinical Implications

Sex-differential patterns reveal drug–target–adverse event pathways with measurable sex bias. Female patients on HDAC inhibitors warrant enhanced monitoring for infections/metabolic disturbances (3–5× female-higher rates). Male patients on sodium channel inhibitors need male-specific monitoring. SexDiffKG enables sex-stratified risk assessment beyond population-level safety data, supporting precision medicine.

4.4 Limitations

FAERS bias: Female-to-male ratio 1.51 exceeds population drug usage (~1.1–1.2); reflects real rates, differential reporting, medication access patterns. ROR disproportionality measure; does not establish causation. **Embedding:** MRR 0.048 reflects 126K entities, imbalanced relations. **Drug resolution:** 15.2% ChEMBL coverage; does not affect core sex-differential signal detection. **RotatE:** Limited to 25 CPU epochs; GPU training needed for convergence. **Static:** Q4 2024 snapshot; no temporal analysis. **Sex framework:** Binary classification; future work needed for transgender/hormone therapy contexts.

4.5 Future Directions

GPU-accelerated RotatE/ComplEx/TransE (≥100 epochs); temporal signal evolution; dose–response integration; causal inference (BCPNN, Information Component); RxNorm/UMLS drug resolution; graph neural networks (R-GCN, CompGCN); System Organ Class stratification; prospective validation against RCTs and EHR outcomes. Integration of sex hormone biomarkers, CYP polymorphisms, tissue-specific expression atlases will further refine predictions.

4.6 Conclusions

SexDiffKG advances computational pharmacovigilance by providing the first systematic, molecularly-integrated resource for sex-differential drug safety. Identification of 49,026 strong signals, distinct clustering into safety profiles, and linking to 429 gene targets with sex-biased patterns provides new insights. Validation (67% confirmation) and audit (85 PASSED) demonstrate quality. Complete public availability of code, data, and documentation enables research and clinical communities to advance sex-aware drug safety assessment and precision medicine.

5. Data Availability

All computation on NVIDIA DGX Spark (Grace Blackwell GB10, ARM64, 128GB). Complete dataset deposited on Zenodo (CC-BY 4.0): https://doi.org/10.5281/zenodo.18819192. Source code (Python 3.13, 45 scripts) at GitHub (GPL-3.0): https://github.com/jshaik369/SexDiffKG. Artifacts: nodes.tsv (127,064 rows), edges.tsv (5,839,718 rows), embeddings, sex-bias scores, cluster profiles, signals, audit results. All scripts deterministic with fixed random seeds. Runtime on GB10: ~15h wall-clock (data 2h, ROR 1h, DistMult 3.5h, RotatE 7.5h, post-processing 1h).

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